

The Spiderweb Action v.3

October 6, 2026

Derivation of the action

The euclidian action for any of the two is derived by trotter expanding the partition function $Z = \text{Tr}(\exp(-\beta H))$ while imposing the Gauss-law constraints $G(E) = 0$. The constraint can be imposed by injecting a projector into the trotter expansion $\text{Tr}(\exp(-\beta H)P)$ where

$$P = \int \mathcal{D}[\phi] \exp \left(i \sum_{\mathbf{x}} G(\mathbf{x}) \phi(\mathbf{x}) \right) \quad (1)$$

This is effectively analogous to the delta function $\delta(f(x)) = \int dy \exp(ify)$ that forces $f(x)$ to be zero under any integral by introducing the auxiliary variable y . I begin with the QED3 case. The Hamiltonian is

$$H_{\text{QED3}} = \frac{U}{2} \sum_{\mathbf{x}} E(\mathbf{x})^2 - K \sum_{\mathbf{x}} \cos(B(\mathbf{x})) \quad (2)$$

Next one performs the trotter expansion

$$Z = \text{Tr} \left[\left(\exp \left(-\frac{\beta}{N} H \right) P \right)^N \right] \quad (3)$$

$$= \int \mathcal{D}[A] \prod_{\tau=1}^N \langle A_{\tau} | \exp(-\epsilon H) P | A_{\tau+1} \rangle \quad (4)$$

This is how [1] writes it. Next the exponential factor is split open into a part only containing the vector potential and a part only containing the electric fields

$$\exp(-\epsilon H) \simeq \exp(-\epsilon H_A) \exp(-\epsilon H_E)$$

This is only an approximation because E and A do not commute (they are conjugate variables). At small ϵ terms $O(\epsilon^2)$ which would contain such commutators can be ignored. Next an eigenbasis of E-fields is inserted

$$Z \simeq \int \mathcal{D}[A] \prod_{\tau=1}^N \sum_{E_{\tau}} \langle A_{\tau}(\mathbf{x}) | \exp(-\epsilon H_A) | E_{\tau}(\mathbf{x}) \rangle \langle E_{\tau}(\mathbf{x}) | \exp(-\epsilon H_E) | A_{\tau+1}(\mathbf{x}) \rangle \quad (5)$$

I do not really understand, why [1] writes $\sum_{E_{\tau}}$ and not an integral. Maybe because he assumes that the E-field takes integer values?

Since E and A are conjugate fields the inner product of their states will simply be of the form

$$\langle A_{\tau} | E_{\tau} \rangle = \exp \left(i \sum_{\mathbf{x}} A_{\tau}(\mathbf{x}) E_{\tau}(\mathbf{x}) \right) \quad (6)$$

and with P written out explicitly, the partition function becomes

$$Z \simeq \iint \mathcal{D}[A] \mathcal{D}[\phi] \sum_{E_\tau} \prod_{\tau=1}^N \exp \left(-\epsilon H_A - \epsilon H_E - i \sum_{\mathbf{x}} G(E_\tau(\mathbf{x})) \phi_\tau(\mathbf{x}) + i \sum_{\mathbf{x}} (A_\tau(\mathbf{x}) E_\tau(\mathbf{x}) - A_{\tau+1}(\mathbf{x}) E_\tau(\mathbf{x})) \right) \quad (7)$$

In the last step the Gauss-law is evaluated and shifted onto ϕ via partial integration. In the case of QED3

$$\sum_{\mathbf{x}} G(E(\mathbf{x})) \phi(\mathbf{x}) = \sum_{\mathbf{x}} (\nabla \cdot E(\mathbf{x})) \phi(\mathbf{x}) = - \sum_{\mathbf{x}} E^i(\mathbf{x}) \partial_i \phi(\mathbf{x})$$

Under the redefinition $\phi_\tau \equiv A_\tau^0$ the full partition function becomes

$$Z \simeq \iint \mathcal{D}[A] \mathcal{D}[\phi] \sum_{E_\tau} \prod_{\tau=1}^N \exp \left(\epsilon K \sum_{\mathbf{x}} \cos(\nabla \times A_\tau) - \epsilon \frac{U}{2} \sum_{\mathbf{x}} E_\tau(\mathbf{x})^2 + i \sum_{\mathbf{x}} E_\tau^i(\mathbf{x}) \partial_i A_\tau^0(\mathbf{x}) + i \sum_{\mathbf{x}} (A_\tau(\mathbf{x}) E_\tau(\mathbf{x}) - A_{\tau+1}(\mathbf{x}) E_\tau(\mathbf{x})) \right)$$

The electric field can be "integrated out" via

$$\exp(z \cos(\phi)) \simeq \sum_n \exp \left(-\frac{1}{2z} n^2 + i \phi n \right)$$

(see equation (A7) in [1]). The action becomes

$$Z \simeq \int \mathcal{D}[A] \prod_{\tau=1}^N \exp \left(\epsilon K \sum_{\mathbf{x}} \cos(\nabla \times A_\tau) + \frac{1}{\epsilon U} \sum_{\mathbf{x}} \cos(\partial_i A_\tau^0(\mathbf{x}) - \partial_\tau A_\tau(\mathbf{x})) \right) \quad (8)$$

The euclidean action is

$$S_E = -\epsilon K \sum_{\mathbf{x}} \cos(\nabla \times A_\tau) - \frac{1}{\epsilon U} \sum_{\mathbf{x}} \cos(\partial_i A_\tau^0(\mathbf{x}) - \partial_\tau A_\tau(\mathbf{x})) \quad (9)$$

Crucially both terms have a minus sign, as opposed to the familiar $L \sim E^2 - B^2$ from standard electrodynamics. This is because we are working in imaginary time and one such imaginary time step sits in the denominator of a fraction. In other words, switching to real time would endow the electric field term with an additional minus sign.

Now to the spiderweb model. The fields live on a square lattice with two basis sites ("crossed" $\boxtimes$ and "empty" $\square$). The electric field $E_i^{\mu\nu}$ is a symmetric traceless tensor. Its two inequivalent components E_i^{xy} and E_i^{xx} live on the crossed $\boxtimes$ and empty $\square$ sites

respectively. The electric field is directly related to the z component of the underlying spin field S_i^z and hence the constrainer can be written as

$$\begin{aligned} C_{\boxtimes} &= E_{\boxtimes_1}^{xx} + E_{\boxtimes_2}^{xy} - E_{\boxtimes_3}^{xx} - E_{\boxtimes_4}^{xy} + E_{\boxtimes_5}^{xx} + E_{\boxtimes_6}^{xy} - E_{\boxtimes_7}^{xx} - E_{\boxtimes_8}^{xy} \\ &= (E_{\boxtimes_1}^{xx} - E_{\boxtimes_3}^{xx} + E_{\boxtimes_5}^{xx} - E_{\boxtimes_7}^{xx}) + (E_{\boxtimes_2}^{xy} - E_{\boxtimes_4}^{xy} + E_{\boxtimes_6}^{xy} - E_{\boxtimes_8}^{xy}) \\ &= D_{xx}E^{xx} + D_{xy}E^{xy} \end{aligned} \quad (10)$$

The magnetic field is defined similarly although only living on empty $\square$ sites. It is defined as

$$\begin{aligned} B &= A_{\square_1}^{xy} - A_{\square_2}^{xx} - A_{\square_3}^{xy} + A_{\square_4}^{xx} + A_{\square_5}^{xy} - A_{\square_6}^{xx} - A_{\square_7}^{xy} + A_{\square_8}^{xx} \\ &= (A_{\square_1}^{xy} - A_{\square_3}^{xy} + A_{\square_5}^{xy} - A_{\square_7}^{xy}) - (A_{\square_2}^{xx} - A_{\square_4}^{xx} + A_{\square_6}^{xx} - A_{\square_8}^{xx}) \\ &= D_{xx}A^{xy} - D_{xy}A^{xx} \end{aligned} \quad (11)$$

The A^{xy} and A^{xx} fields live, similarly to E on the crossed $\boxtimes$ and empty $\square$ plaquettes respectively. Structurally, 11 looks like a rotated version of 10.

Next the same steps as in QED3 can be taken to arrive at the euclidean spiderweb action. Up until equation 7 nothing fundamentally changes other than the fact, that "products" between A and E are now traces over matrix products. Explicitly

$$Z \simeq \iint \mathcal{D}[A]\mathcal{D}[\phi] \sum_{E_\tau} \prod_{\tau=1}^N \exp \left(-\epsilon H_A - \epsilon H_E - i \sum_{\boxtimes_i} D_{xy}(E_\tau^{xy}(\mathbf{r}_i))\phi_\tau(\mathbf{r}_i) - \right. \quad (12)$$

$$\left. i \sum_{\square_i} D_{xx}(E_\tau^{xx}(\mathbf{r}_i))\phi_\tau(\mathbf{r}_i) + i \sum_{\mathbf{r}_i} \left(A_\tau^{ij}(\mathbf{r}_i)E_\tau^{ij}(\mathbf{r}_i) - A_{\tau+1}^{ij}(\mathbf{r}_i)E_\tau^{ij}(\mathbf{r}_i) \right) \right) \quad (13)$$

The last sum $\sum_i$ should be understood as summing over all sites. The last sum is understood as summing over all sites, where A and E select their components based on the sublattice $\mathbf{r}_i$ sits on.

One now has to perform partial integration to "shift the Gauss law" onto A^0 . This can be easily done on the lattice. Since the sum $\sum_{\boxtimes}$ runs over all boxes, one can rewrite it as

$$\sum_{\boxtimes} (E_{\boxtimes_2}^{xy} - E_{\boxtimes_4}^{xy} + E_{\boxtimes_6}^{xy} - E_{\boxtimes_8}^{xy}) A_{\boxtimes_0}^0 = \sum_{\boxtimes} E_{\boxtimes_0}^{xy} (A_{\boxtimes_2}^0 - A_{\boxtimes_4}^0 + A_{\boxtimes_6}^0 - A_{\boxtimes_8}^0) \quad (14)$$

where $\boxtimes_0$ labels the center crossed box. The same trivial rewriting can be done for the $\square$ sum. The partition function now becomes

$$\begin{aligned} Z \simeq \iint \mathcal{D}[A]\mathcal{D}[\phi] \sum_{E_\tau} \prod_{\tau=1}^N \exp \left(\epsilon K \sum_{\square} \cos(A_{\square_1}^{xy} - A_{\square_2}^{xy} + \dots) - \epsilon \frac{U}{2} \sum_{\mathbf{r}_i} (E_\tau^{ij}(\mathbf{x}))^2 \right. \\ - i \sum_{\boxtimes_i} E_{\boxtimes,\tau}^{xy}(\mathbf{r}_i) (A_{\tau,\boxtimes_2}^0 - A_{\tau,\boxtimes_4}^0 + A_{\tau,\boxtimes_6}^0 - A_{\tau,\boxtimes_8}^0) \\ - i \sum_{\square_i} E_{\square,\tau}^{xx}(\mathbf{r}_i) (A_{\tau,\square_1}^0 - A_{\tau,\square_3}^0 + A_{\tau,\square_5}^0 - A_{\tau,\square_7}^0) \\ \left. + i \sum_{\mathbf{r}_i} \left(A_\tau^{ij}(\mathbf{r}_i)E_\tau^{ij}(\mathbf{r}_i) - A_{\tau+1}^{ij}(\mathbf{r}_i)E_\tau^{ij}(\mathbf{r}_i) \right) \right) \end{aligned} \quad (15)$$

At this point one can again use the identity

$$\exp(z \cos(\phi)) = \exp\left(\sum_n -\frac{n^2}{2z} + in\phi\right)$$

where n is taken to be the electric field.

$$\begin{aligned} \exp\left(-i \sum_{\boxtimes} E_{\boxtimes,\tau}^{xy} (D_{xy}(A_\tau^0) - \partial_\tau A_{\boxtimes,\tau}^{xy}) - i \sum_{\square} E_{\square,\tau}^{xx} (D_{xx}(A_\tau^0) - \partial_\tau A_{\square,\tau}^{xx}) - \epsilon \frac{U}{2} \sum_{\mathbf{r}_i} E_{ij}^{ij} E_{ij}\right) \\ = \exp\left(\frac{1}{\epsilon U} \cos(D_{xy}(A_\tau^0) - \partial_\tau A_{\boxtimes,\tau}^{xy})\right) \cdot \exp\left(\frac{1}{\epsilon U} \cos(D_{xx}(A_\tau^0) - \partial_\tau A_{\square,\tau}^{xx})\right) \end{aligned} \quad (16)$$

Inserting this into the partition function finally gives for the euclidean action

$$\begin{aligned} S \simeq -\epsilon K \sum_{\square} \cos(A_{\square_1}^{xy} - A_{\square_2}^{xx} - A_{\square_3}^{xy} + A_{\square_4}^{xx} + A_{\square_5}^{xy} - A_{\square_6}^{xx} - A_{\square_7}^{xy} + A_{\square_8}^{xx}) \\ - \frac{1}{\epsilon U} \sum_{\boxtimes} \cos(A_{\boxtimes_2}^0 - A_{\boxtimes_4}^0 + A_{\boxtimes_6}^0 - A_{\boxtimes_8}^0 - A_{\boxtimes,\tau+1}^{xy} + A_{\boxtimes,\tau}^{xy}) \\ - \frac{1}{\epsilon U} \sum_{\square} \cos(A_{\square_1}^0 - A_{\square_3}^0 + A_{\square_5}^0 - A_{\square_7}^0 - A_{\tau+1}^{xx} + A_{\tau}^{xx}) \end{aligned} \quad (17)$$

The three cosines contain respectively B , E^{xy} and E^{xx} . They are shown in figure 2. The A^0 fields consistently live on the links connecting crossed squares. The magnetic terms are temporarily disconnected while the electric terms connect fields of different time-slices.

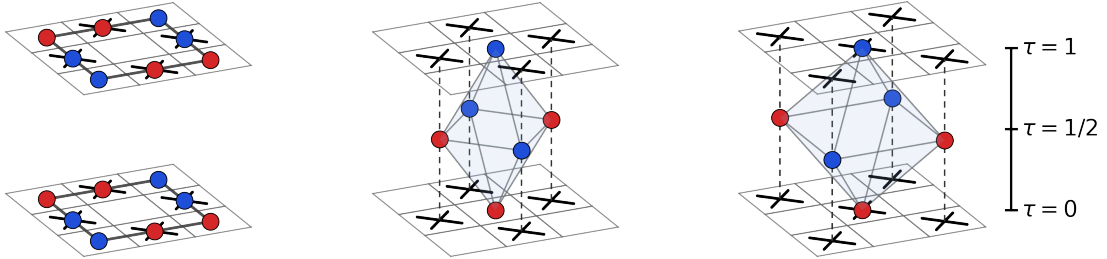


Figure 1: Operators appearing in the cosines of eq 17. Left: Two disconnected B -terms living on two different time slices. Middle: E^{xx} -term: Two time-slices are connected by A^0 -fields living on the links between crossed sites. Right: E^{xy} -terms.

Some plots

The specific heat shows a peak but no signs of finite size scaling. At low temperatures The cosines eq. 17 can be expanded around the ground state giving a harmonic theory. Each independent degree of freedom contributes $k_B T/2$ to the specific heat. Counting

the A^0 -links as sites as well one has N A -fields in total of which one has to remove gauge transformations

$$A^0 \rightarrow \Delta_\tau f \quad (18)$$

$$A^{ij} \rightarrow D^{ij} f \quad (19)$$

By the definition of the D^{ij} one can see that f is only defined on empty squares, meaning $N/3$ of all sites. A constant f does not change any of the A s, so gauge transformations remove a total of $N/3 - 1$ degrees of freedom. There are also f s built from the functions x , y or $x^2 + y^2$ that are annihilated by D_{xx} and D_{xy} but they are not periodic and hence do not contribute. In addition to the gauge transformations there are also constant shifts of the A -fields that don't contribute to the energy, removing additional 3 degrees of freedom. In total there are hence $N - (N/3 - 1) - 3 = 2N/3 - 2$ independent degrees of freedom each contributing $k_B T/2$ in energy giving a specific heat per site of $c_v \simeq k_B T/3$.

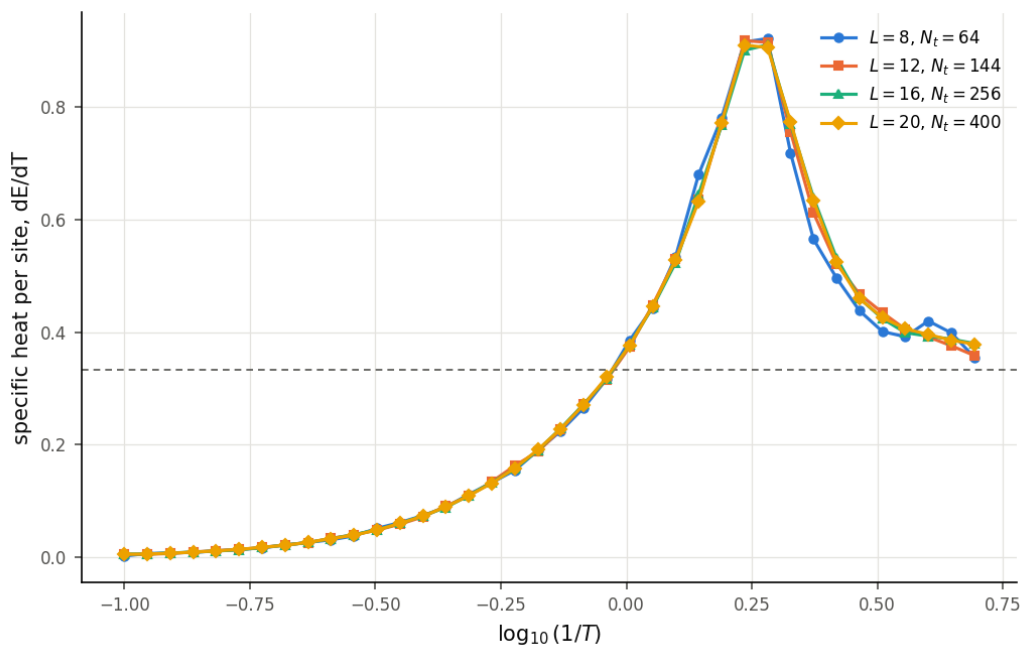


Figure 2: Specific heat obtained from the temperature derivative of the mean energy.

It is possible to define a helicity modulus for lattice U(1) theories [3] by inserting a global flux into the action, so it should also be possible here. Setting $B \rightarrow B + \alpha$ in the

action and computing the second derivative of the partition function gives

$$\begin{aligned}
\partial_\alpha \ln Z &= \frac{1}{Z} \partial_\alpha \int \mathcal{D}[A] \exp \left(\beta \left(\sum_{\square} \cos(B_{\square} + \alpha) + \frac{1}{KU} (\text{E-terms...}) \right) \right) \\
&= \left\langle -\beta \sum_{\square} \sin(B_{\square} + \alpha) \right\rangle \\
\partial_\alpha^2 \ln Z &= - \left\langle -\beta \sum_{\square} \sin(B_{\square} + \alpha) \right\rangle^2 + \left\langle \beta^2 \left(\sum_{\square} \sin(B_{\square} + \alpha) \right)^2 \right\rangle - \left\langle \beta \sum_{\square} \cos(B_{\square} + \alpha) \right\rangle
\end{aligned} \tag{20}$$

Since the euclidean action is even under $B \rightarrow -B$ or equivalently $A \rightarrow -A$, the first term vanishes and the magnetic helicity modulus is

$$\gamma_B = \left\langle \sum_{\square} \cos(B_{\square}) \right\rangle - \frac{1}{T} \left\langle \left(\sum_{\square} \sin(B_{\square}) \right)^2 \right\rangle \tag{21}$$

Its plot is shown in figure 3. The helicity modulus does not look clean when simulated starting from a hot state. Figure 3 is instead simulated from heating up from the ground state. A more detailed run needs to be performed on the cluster to gain more insight, however the plot here already seems to indicate a sort of transition around $T \approx 0.5 - 0.55$.

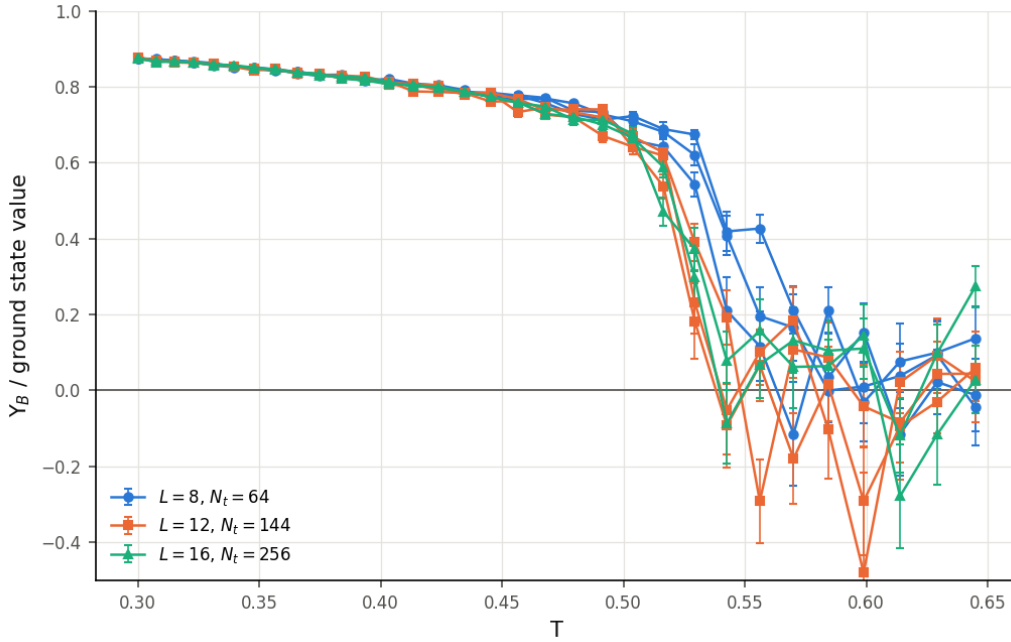


Figure 3: Helicity modulus obtained by heating from the ground state.

Instantons?

Before discretizing, I believe, one can make a statement on the continuum-side that will be useful for detecting instantons later. I start with QED3. The action is

$$S_E = -\beta \sum_i (\cos(B) + \cos(E^x) + \cos(E^y)) \quad (22)$$

This is effectively just electrostatics on a generalized magnetic field $\mathbf{B}$

$$\mathbf{B} = (B, E^x, E^y) \quad (23)$$

So its clear that the instanton will be a divergence on $\mathbf{B}$. In a non-compact theory this vanishes

$$\nabla \cdot \mathbf{B} = \partial_0 \phi_{xy} + \partial_y \phi_{0x} + \partial_x \phi_{y0} = 0 \quad (24)$$

with ϕ_{ij} the flux through the ij plaquette. However in a compact theory the fluxes ϕ_{ij} are only defined up to multiples of 2π and hence the divergence of $\mathbf{B}$ must be calculated from the *wrapped* fluxes

$$\begin{aligned} \nabla \cdot \mathbf{B}(r) &= \partial_0 \bar{\phi}_{xy} + \partial_x \bar{\phi}_{y0} + \partial_y \bar{\phi}_{0x} \\ &= \partial_0 (\phi_{xy} - [\phi_{xy}]_{2\pi}) + \dots \\ &= \partial_0 [\phi_{xy}(r)]_{2\pi} + \partial_x [\phi_{y0}(r)]_{2\pi} + \partial_y [\phi_{0x}(r)]_{2\pi} \end{aligned} \quad (25)$$

with $[\phi_{ij}]_{2\pi}$ referring to the flux rounded to a multiple of 2π . The divergence-free condition $\nabla \cdot \mathbf{B}$ in the non-compact theory is key to finding the instantons. If one didnt know where to look for instantons, one could instead look for an equation like eq. 24, discretize it and wrap every term. For example in QED3, if one discretizes derivatives as $\partial_x f(r) = f(r+x) - f(r)$ then it immediately follows that

$$\begin{aligned} \nabla \cdot \mathbf{B}(r) &= [A_y(r+x+\tau) - A_y(r+\tau) - A_x(r+y+\tau) + A_x(r+\tau)]_{2\pi} \\ &\quad - [A_y(r+x) - A_y(r) - A_x(r+y) + A_x(r)]_{2\pi} + \dots \end{aligned}$$

One can immediately see that the wrapped fluxes live on plaquettes and that all fluxes together live on a cubic surface, however one does not need to know. The only thing that happend was 1. finding a divergence law that is identically zero in the non-compact theory, 2. wrap its individual terms to the interval $[0, 2\pi)$ and 3. discretize the derivatives.

The spiderweb model is already discretized so one only needs to find an equation that vanishes in the non-compact version. This equation is

$$\Delta_t B + D_{xx} E^{xy} - D_{xy} E^{xx} = 0 \quad (26)$$

Where

$$\Delta_t B = B(\mathbf{x} + \tau) - B(\mathbf{x}) \quad (27)$$

$$D_{xx} E^{xy} = E^{xy}(\mathbf{x} + x) - E^{xy}(\mathbf{x} + y) + E^{xy}(\mathbf{x} - x) - E^{xy}(\mathbf{x} - y) \quad (28)$$

$$D_{xy} E^{xx} = E^{xx}(\mathbf{x} + x + y) - E^{xx}(\mathbf{x} - x + y) + E^{xx}(\mathbf{x} - x - y) - E^{xx}(\mathbf{x} + x - y) \quad (29)$$

Importantly this equation only holds on empty squares $\square$. This equation is easily proven from the definitions of B and E :

$$\begin{aligned}\Delta_t B + D_{xx} E^{xy} - D_{xy} E^{xx} &= \Delta_t (D_{xx} A^{xy} - D_{xy} A^{xx}) \\ &\quad + D_{xx} (D^{xy} A^0 - \Delta_t A^{xy}) \\ &\quad - D_{xy} (D^{xx} A^0 - \Delta_t A^{xx}) \\ &= 0\end{aligned}\tag{30}$$

To find instantons one can now simply wrap the individual terms to the interval $[0, 2\pi)$ and only sum up the integer windings

$$n_{\text{inst}} = \Delta_t [B]_{2\pi} + D_{xx} [E^{xy}]_{2\pi} - D_{xy} [E^{xx}]_{2\pi}\tag{31}$$

where again $[\cdot]$ is the integer winding number around 2π or the integer division by 2π .

```
std::vector<int> Spiderweb_corrected::getInstantons() const
{
    // Get B and E fields at every site wrapped to the interval [-1,1]
    std::vector<double> b(nSites, 0.0), e(nSites, 0.0);
    for (int site = 0; site < nSites; site++)
    {
        if (isType1(site))
            e[site] = mcf::mapToCircle(electricField_xy(site));
        else
        {
            b[site] = mcf::mapToCircle(magneticField(site));
            e[site] = mcf::mapToCircle(electricField_xx(site));
        }
    }

    // Instantons are computed as violations of the gauss law Dt B + Dxx E^xy - Dxy E^xx = 0
    std::vector<int> instantons(nSites, 0);
    for (int r = 0; r < nSites; r++)
    {
        if (isType1(r))
            continue;

        const double wrapped_gauss = b[shifted(r, 0, 0, +1)] - b[r]
            // Dxx [E^xy]: the four type 1 neighbours
            + e[shifted(r, +1, 0, 0)] - e[shifted(r, 0, +1, 0)] + e[shifted(r, -1, 0, 0)] - e[shifted(r, 0, -1, 0)]
            // - Dxy [E^xx]: the four diagonal type 2 neighbours
            - (e[shifted(r, +1, +1, 0)] - e[shifted(r, -1, +1, 0)] + e[shifted(r, -1, -1, 0)] - e[shifted(r, +1, -1, 0)]);
        instantons[r] = (int)std::lround(wrapped_gauss / 2.0);
    }
    return instantons;
}
```

Figure 4: code implementation of the instantons: First, all magnetic and electric fluxes are wrapped. The values of the wrapped fluxes are then inputted into the Gauss law. Violations are stored in an array. Because I calculate the energy as $\cos(\pi \cdot \text{Fields})$, all fields have period 2 and I hence need to divide the result by two to get the correct integer.

The energy per site shows a sort of smooth jump around $T = 0.55$ as well as the instanton density. The energy density looks similar to the transition seen in [2] but there the transition was in the chemical potential, so this is likely just a coincidence.

Visualizing the defects obtained as outlined above shows in general two kinds of defects (see Figure 6): Immobile single defects and partially mobile double defects. These are very likely simply the fractonic charges and subdimensionally mobile dipoles. At times one can even observe spontaneous excitation of quadrupoles, transforming a single charge into three charges and/or the opposite, which is exactly what one expects.

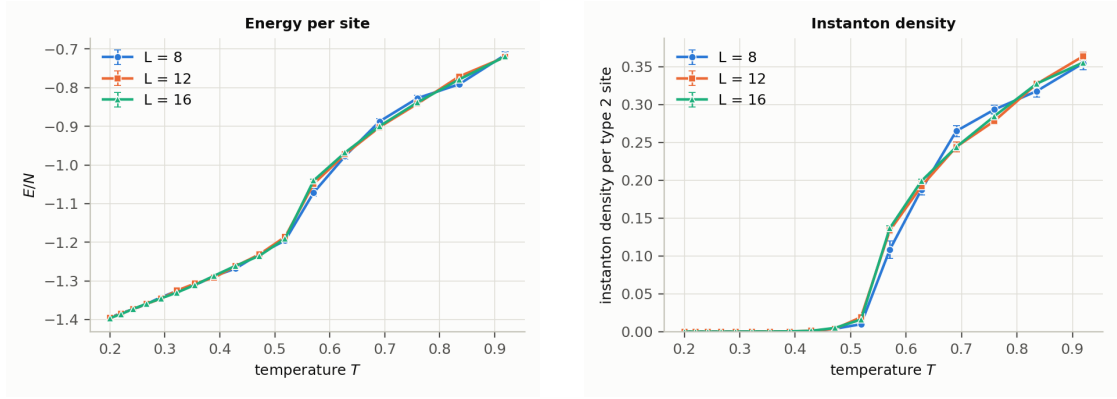


Figure 5: Left: Energy density per site, showing a kink around $T = 0.55$. Right: "Instanton" density showing a sharp increase at around $T = 0.55$. Both data was obtained from a heating simulation.

Whether or not there are also temporal defects that one could call instantons is still an open question.

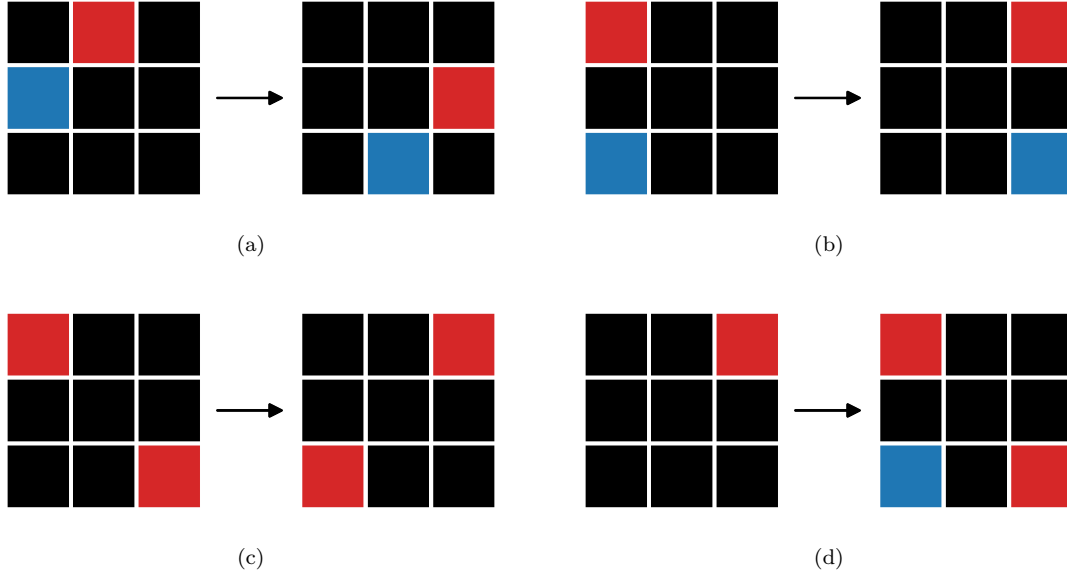


Figure 6: Excitations observed visually from the simulation: (a): A dipole moving diagonally. (b) a dipole moving horizontally. (c) two seemingly isolated charges flipping along an axis due to the action of a quadrupole. (d) an isolated charge transforming into three charges due to the action of a quadrupole

References

- [1] José M Cruz et al. “Hollow lattice tensor gauge theories with bosonic matter”. In: *Physical Review D* 112.11 (2025), p. 114508.
- [2] Nils Niggemann et al. “Gapless fracton quantum spin liquid and emergent photons in a 2D spin-1 model”. In: *Nature Communications* (2026).
- [3] Michele Vettorazzo and Philippe De Forcrand. “Electromagnetic fluxes, monopoles, and the order of 4d compact U (1) phase transition”. In: *Nuclear Physics B* 686.1-2 (2004), pp. 85–118.